Running Data Accuracy: Comparing Upper-Arm Sensors and Smartwatches

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Abstract

The accurate tracking of heart rate (HR) and geospatial data during sports activities is crucial for reliable performance analysis. Although wrist-worn smartwatches are considered the consumer standard, physical motion artifacts often compromise data integrity. This paper compares the data continuity and cross-device agreement of a wrist-worn smartwatch (Amazfit GTR 3 Pro) and an optical upper-arm sensor (Coospo HW9) during outdoor running. Because both devices are consumer-grade and neither is a clinically validated reference (e.g., an ECG chest strap), our results describe *agreement between two consumer sensors* rather than absolute accuracy against ground truth. Based on a synchronized field study ($N = 6$ trials with a single runner), average per-run HR means align closely between devices, but instantaneous agreement is more variable: aggregated across all six runs, mean Pearson $r = 0.60$ (median 0.66), mean MAE $= 8.9$ bpm, and mean bias $= +4.6$ bpm. Geospatial tracks agree reasonably well overall (mean ΔGPS ≈ 4.56 m across all six runs, ranging from 2.1 to 8.3 m per-run mean, max. 30.5 m; mean track-length error $\approx 1.5\%$, range 0.5–4.2%). Speed agreement falls between these two: mean Pearson $r = 0.73$ (median 0.75) and mean MAE $= 1.09$ km/h across all six runs. The wrist-worn device shows substantial signal dropouts, missing heart rate data for up to 74.6% of a run's duration and failing to register several transient cardiac load peaks. A post-hoc power analysis indicates that $n \approx 13$ trials would be required for a conventionally powered validation; at $N = 6$ trials from a single subject, this study reaches only 53.2% of that target and should be read as an underpowered, exploratory pilot rather than a confirmatory validation. We further note that the two devices differ not only in sensor placement but also in hardware generation and physical condition, so placement effects cannot be cleanly separated from these confounds in the present design.

1 Introduction

Wearable fitness trackers have democratized performance diagnostics for recreational and professional athletes alike. For runners, real-time heart rate (HR) monitoring serves as a primary metric for training intensity distribution and metabolic threshold management. However, the commercial market is split between different form factors and sensor placements.

Wrist-worn smartwatches leverage photoplethysmography (PPG) to measure blood volume changes through the skin [Kim and Baek 2023]. While highly convenient, this placement is notoriously susceptible to motion-induced artifacts caused by rhythmic arm swings, gripping forces, and rapid cadence shifts during running [Kim and Baek 2023; Schweizer and Gilgen-Ammann 2024]. These artifacts introduce high-frequency noise, often leading to signal loss or algorithmic miscalculations. Alternatively, upper-arm mounted sensors present a promising compromise, placing the optical array on a larger muscular surface (e.g., biceps brachii) with less dynamic independent movement [Schweizer and Gilgen-Ammann 2024].

This paper compares the tracking behavior, signal reliability, and geospatial mapping of a wrist-worn smartwatch and an upper-arm mounted sensor. Because the upper-arm device is itself a consumer sensor rather than a validated reference standard, we frame our contribution as a quantification of *cross-device agreement and data density*, not as an absolute accuracy assessment. By evaluating data density, velocity profiles, and peak-value detection across a synchronized field study, we document the operational limitations of the wrist-based device under the conditions tested here.

2 State of the Art

Validation studies for consumer wearables generally fall into two categories: controlled laboratory environments using treadmill protocols, and free-living field tests. Wearable GPS-enabled sport watches and smartphones are now widely adopted among amateur athletes for tracking and optimizing training [Pobiruchin et al. 2017].

2.1 Wrist-Based PPG Limitations

Prior research indicates that wrist-worn devices demonstrate acceptable accuracy during low-motion states (e.g., cycling or walking). Commercial wearables are generally accurate for measuring heart rate in laboratory settings, with devices from established brands performing best [Fuller et al. 2020]. However, none of the wrist-worn devices tested in these reviews have proven accurate for measuring energy expenditure [Fuller et al. 2020; Germini et al. 2022]. The mechanical displacement of the watch body creates transient air gaps between the photodiode and the skin, causing ambient light leakage [Kim and Baek 2023]. Although green light is widely used in wearable PPG for its absorption characteristics, it remains vulnerable to motion artifacts during vigorous activity [Kim and Baek 2023]. During triathlon training, wrist smartwatches have been shown to exhibit mean absolute percentage errors for heart rate between 3.1% and 8.3%, with distance-tracking accuracy dropping further during complex motion [Jacko et al. 2024].

2.2 Alternative Sensor Placements

To mitigate these limitations, sports science literature has explored alternative body sites. Optical sensors worn on the forearm or upper arm benefit from a more stable tissue mass with less independent motion. Analytical validation studies show that upper-arm optical sensors exhibit good validity and reliability, whereas wrist-worn devices show accuracy that varies strongly with the level of motion artifact [Schweizer and Gilgen-Ammann 2024]. Beyond the wrist-versus-upper-arm question addressed here, in-ear PPG has been validated as a further alternative placement with good agreement against ECG-derived heart rate [**?**], chest-strap sensors remain the most extensively validated consumer option for cardiac monitoring despite known motion and skin-contact limitations [**?**], and multi-sensor fusion approaches that combine several imperfect signal sources have been proposed as a way to improve robustness beyond what any single placement can achieve [**?**]. It should be noted that such comparisons, including our own, typically evaluate agreement between two consumer-grade devices rather than accuracy against a clinical reference such as an electrocardiogram (ECG) chest strap.

3 Methodology

To evaluate the systematic differences between the two tracking topologies, a synchronized field testing protocol was established. The study was conducted on an asphalted outdoor route characterized by a dense tree canopy, providing a challenging environment for satellite-based navigation.

3.1 Hardware Configuration

Two distinct commercial optical devices were worn simultaneously by the same subject:

- **Wrist Device (HG):** Amazfit GTR 3 Pro with a BioTracker PPG 3.0 sensor, sampling at a nominal frequency of 1 Hz. This device had an operational lifespan exceeding four years at the time of testing, with visible mechanical degradation of the optical sensor window (discussed in Section 5).
- **Upper-Arm Device (OA):** Coospo HW9 optical heart rate sensor, attached via an elastic band to the dominant biceps brachii and streaming via Bluetooth LE. This device was newer, both in hardware generation and physical condition.

Because the two devices differ simultaneously in sensor placement, hardware generation, and physical wear, placement effects cannot be isolated from these confounds in the present design; we return to this limitation in Section 5.

3.2 Data Acquisition & Processing

The evaluation uses a data matrix comprising $N = 6$ distinct outdoor running trials, each a single continuous run along the same route, performed by the same subject on separate occasions. An intra-subject design was chosen to eliminate inter-subject demographic confounders. For data synchronization, timestamps from both devices were independently rounded to the nearest full second and joined via an inner merge on this common 1-second key, so that only strictly overlapping, mutually valid samples contributed to the comparison; the resulting synchronized duration corresponds to the intersection of both devices' recording windows. A structural anomaly discovered during preprocessing concerned heart rate data density: while the upper-arm sensor recorded continuous telemetry across all six trials, the smartwatch produced substantial gaps in 4 of the 6 runs, dropping up to 74.6% of cardiac data points, likely attributable to a combination of motion-induced artifacts and the sensor degradation described in Section 5.

4 Results

The empirical analysis highlights a contrast in signal stability rather than a large deviation in baseline geospatial tracks. As compiled in Table 1, mean heart rate calculated across stable intervals is congruent between devices, but severe dropouts limit the practical utility of the wrist sensor's data. As visualized in Figure 1, the upper-arm sensor (blue) reacts immediately to cardiac micro-intensities, whereas the wrist sensor (orange) exhibits a smoothing lag and substantial data omissions.

Figure 2 shows scatterplot and Bland-Altman agreement for Trial 3 specifically, chosen as an illustrative example; aggregate distributional statistics across all six trials are summarized in Table 1 and in the cross-run figures below.

4.1 Sample Size & Power Validation

In contemporary hardware validation literature for optical heart rate sensors, sample sizes typically range from 30 to 50 *participants* [Jacko et al. 2024; Chow and Yang 2020; Stahl et al. 2016] — a figure not directly comparable to our $n \approx 13$ *trials*, since our design uses repeated trials from a single subject rather than multiple participants (see Section 5). To determine the statistical scalability of our $N = 6$ longitudinal trial matrix, a post-hoc power analysis was conducted for a two-sided paired t-test using Equation 1:

$$n = \frac{(Z_{1-\alpha/2} + Z_{1-\beta})^2 \cdot \sigma_d^2}{\Delta^2} \tag{1}$$

Where $\alpha = 0.05$ ($Z_{0.975} = 1.96$), $1 - \beta = 0.80$ ($Z_{0.80} = 0.84$),

Figure 1: *Comparison of heart rate kinetics (top) and velocity profiles (bottom) during Trial 3. The upper-arm sensor (blue) captures dynamic peaks, whereas the wrist sensor (orange) exhibits latency and data omissions.*

$\Delta = 5$ bpm specifies the targeted minimal detectable difference, and $\sigma_d = 6$ bpm is the standard deviation of differences.

As illustrated in Figure 5(b), solving Equation 1 yields a required sample size of $n \approx 13$ to achieve 80% statistical power. The current sample size of $N = 6$ reaches approximately 53.2% of that target. This study should therefore be read as an **underpowered, exploratory pilot**: it is useful for identifying structural sensor differences worth investigating further, but it is not a confirmatory validation, and effect sizes reported here should not be generalized without a larger, adequately powered, multi-subject replication.

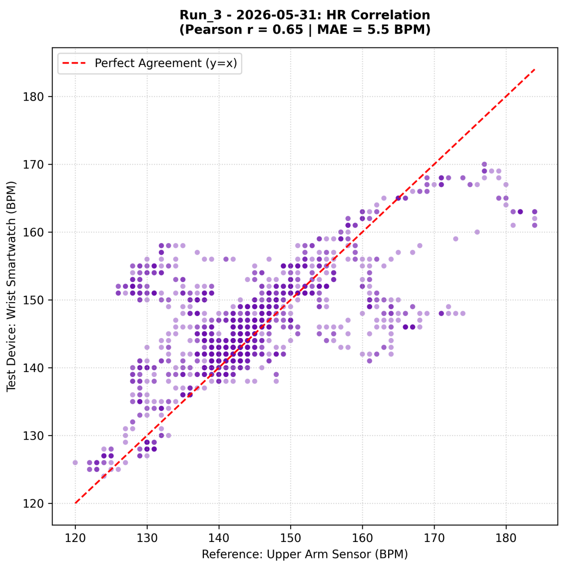
5 Discussion & Limitations

The primary structural vulnerability of the wrist-worn architecture appears to be motion artifacts rather than the optical algorithm itself. Rapid arm accelerations shift the watch body, causing ambient light leakage. This is consistent with the observation that the wrist device's companion app omitted over 1,000 seconds of data in multiple runs, a vulnerability similarly reported for other wrist-worn PPG technologies subjected to arm movement [Schweizer and Gilgen-Ammann 2024]. In Run 2, a *cadence lock* occurred: the wrist strap loosened, causing the device to lock onto the stride rhythm ($\sim$130 bpm) instead of the cardiovascular load ($\sim$150 bpm).

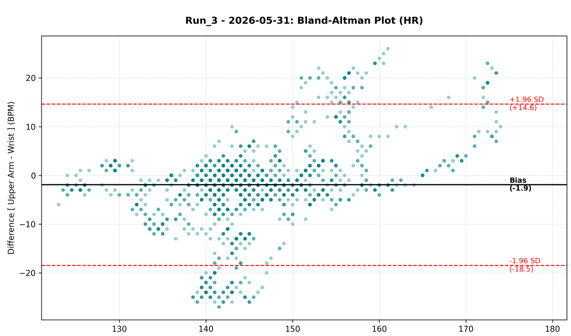
Notably, this single event coincides with Run 2 being an outlier across *several* independently measured dimensions: HR agreement collapses ($r = 0.03$, MAE $= 22.1$ bpm, bias $= +19.9$ bpm), GPS offset peaks (30.5 m, more than $6\times$ the cross-run mean), track-length error is highest (4.2%, roughly $3\times$ the cross-run mean of 1.5%), and the estimated HR time lag (+60 s) sits exactly at the boundary of the $\pm$60 s search window used for the cross-correlation, meaning the true lag may exceed the tested range and the reported value should be treated as a lower bound rather than a precise estimate. The convergence of degraded agreement, offset, and lag metrics in a single run — rather than four independent anomalies scattered across different trials — supports the interpretation that a single mechanical event (the loosened strap) is the common cause, and cautions against averaging Run 2 together with the

Table 1: *Comprehensive sensor and track evaluation ($N = 6$ synchronized runs).*

Metric / Trial	Run 1	Run 2	Run 3	Run 4	Run 5	Run 6	Avg.
Synchronized Duration (s)	1634.0	1556.0	1722.0	1762.0	1617.0	1608.0	1649.8
OA Distance (km)	3.190	2.877	3.071	3.185	3.183	3.213	3.120
HG Distance (km)	3.176	2.911	3.072	3.174	3.168	3.193	3.116
Distance Error (%)	1.1	4.2	2.0	0.5	0.7	0.7	1.5
OA HR Mean / Max (bpm)	148.9/186	150.8/188	144.3/184	145.8/180	142.9/185	153.1/185	147.6/184.7
HG HR Mean / Max (bpm)	149.6/168	130.8/165	146.2/170	142.6/169	144.0/169	155.1/175	144.7/169.3
HG HR Data Loss (s)	1184	29	13	1269	1144	1052	781.8
HR Pearson r / MAE (bpm)	0.83/4.0	0.03/22.1	0.65/5.5	0.60/9.1	0.68/7.1	0.80/5.3	0.60/8.9
Speed Pearson r / MAE (km/h)	0.55/1.65	0.72/1.34	0.77/0.85	0.79/0.76	0.84/0.64	0.72/1.28	0.73/1.09
Mean / Max GPS Offset (m)	3.42/10.5	7.26/30.5	8.26/22.9	3.83/11.1	2.12/6.6	2.48/17.9	4.56/16.6



(a) *Scatterplot (Trial 3, Pearson $r = 0.65$)*



(b) *Bland-Altman plot (Trial 3, Bias $= -1.9$ bpm)*

Figure 2: *Statistical agreement between upper-arm and wrist sensors for Trial 3. The scatterplot (a) reveals a moderate correlation with notable spread. The Bland-Altman plot (b) shows a small mean bias (-1.9 bpm) but wide 95% limits of agreement ($+14.6$ to -18.5 bpm), with underestimation by the wrist sensor increasingly visible at higher heart rates. Trial 3 is close to the cross-run* median *agreement (Figure 3), but not to the cross-run* mean, *which is pulled down by Run 2.*

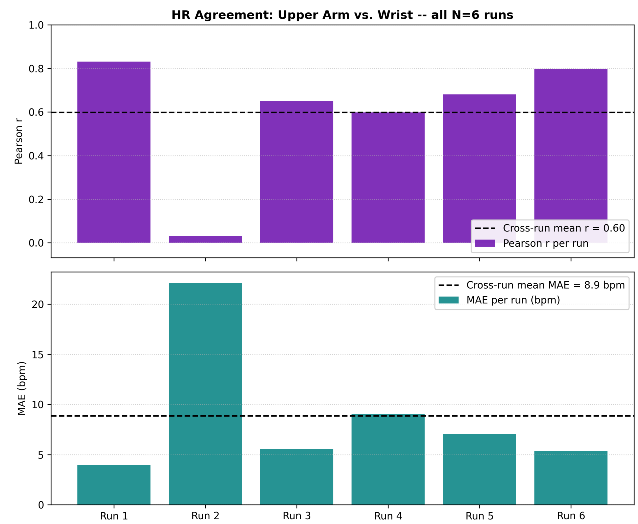


Figure 3: *HR agreement (Pearson r, top; MAE, bottom) between upper-arm and wrist sensors across all $N = 6$ runs. Cross-run mean $r = 0.60$ (median 0.66) and mean MAE $= 8.9$ bpm. The single-trial example in Figure 2 ($r = 0.65$) closely matches the cross-run* median, *but the* mean *is pulled substantially lower by Run 2 ($r = 0.03$, MAE $= 22.1$ bpm), illustrating why a single representative trial can understate the overall variance in device agreement (see Section 5).*

other five runs without flagging it as qualitatively different.

Speed agreement, however, does *not* follow this pattern: Run 2 achieves $r = 0.72$, close to the cross-run mean ($r = 0.73$), while Run 1 — otherwise the cleanest run for HR agreement ($r = 0.83$) — shows the *weakest* speed agreement of all six runs ($r = 0.55$). This dissociation is informative rather than contradictory: PPG-derived HR and GPS-derived position are both sensitive to strap tightness and skin contact, whereas the wrist device's speed estimate can draw on accelerometer/gyroscope data largely independent of optical or GPS quality, plausibly explaining why it degrades on a different run and by a different mechanism. Consequently, the "single confound" account below should be read as applying specifically to the optical and GPS subsystems, not to the device as a whole.

This convergence is not merely visual: treating each run's summary statistic as one observation ($N = 6$), several of these degradation metrics correlate significantly with each other across runs (Table 2). Distance error correlates strongly with both weaker HR agreement

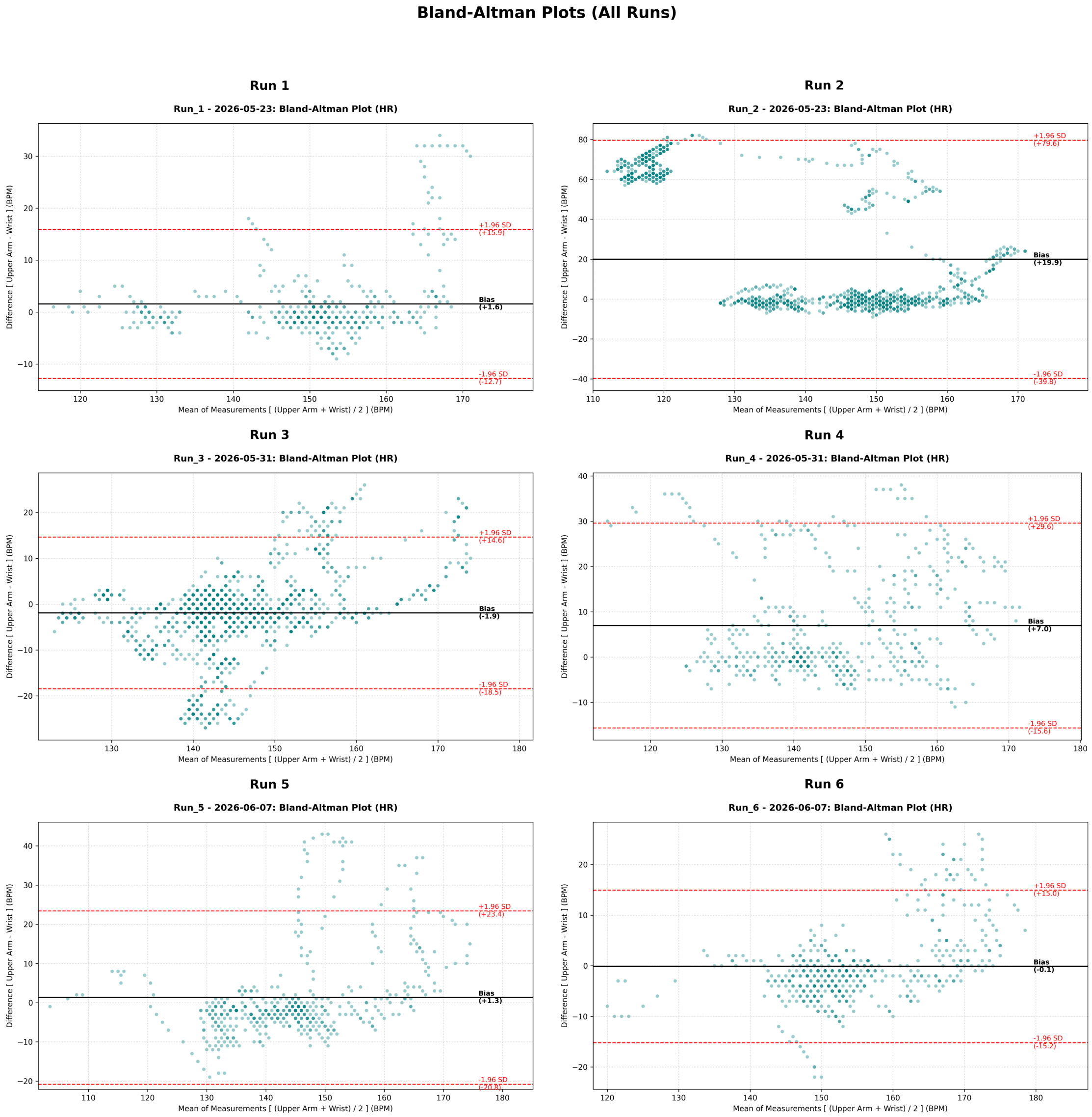
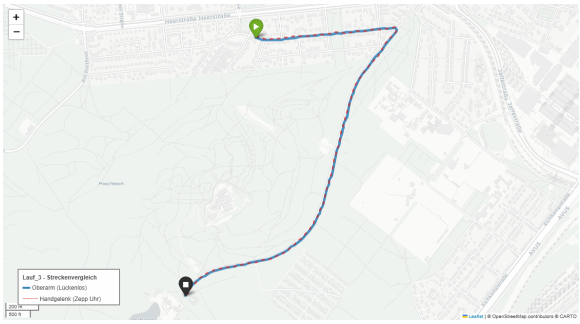


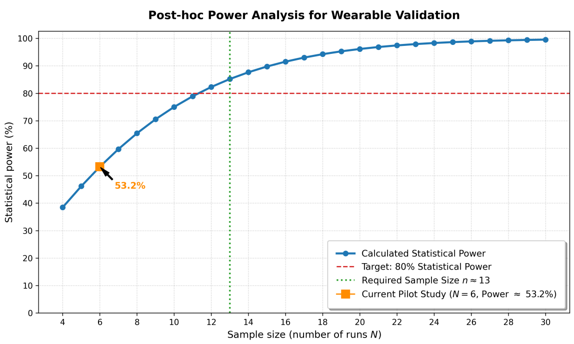
Figure 4: *Bland-Altman plots for all $N = 6$ runs individually. Run 1, 3, 5, and 6 show the expected pattern of tight agreement with modest bias; Run 4 shows a persistent positive bias; Run 2 shows the characteristic three-cluster split described in Section 5, a direct visual signature of the cadence-lock event rather than random measurement noise.*

($r = -0.89$, $p = 0.018$) and higher HR MAE ($r = +0.84$, $p = 0.035$), and HR lag correlates with weaker HR agreement ($r = -0.84$, $p = 0.038$). Speed agreement, by contrast, shows essentially no correlation with any of these metrics ($|r| < 0.25$ in all cases), consistent with the dissociation noted above. Given $N = 6$ these correlations should be read as suggestive rather than confirmatory, but they are consistent with a single latent factor — optical/GPS attachment quality in a given run — driving degradation jointly across the HR and geospatial channels, while speed tracking degrades through a separate, uncorrelated mechanism.

Confounded comparison (device generation and condition). A material limitation of this study is that the two devices differ along more dimensions than sensor placement alone. The Amazfit GTR 3 Pro represents 2021 PPG technology and had, at the time of testing, an operational age of five years with visible mechanical surface degradation — a milky-translucent discoloration of the sensor window that plausibly increases internal light scattering and degrades signal-to-noise ratio. The Coospo HW9, released in June 2024, was new and in optimal physical condition during the study. Consequently, we **cannot cleanly separate the effect of sensor placement (wrist vs. upper arm) from the confounding effects of hardware generation and physical wear**. The 74.6% data-loss figure should therefore be interpreted as characterizing *this specific aged wrist device* under these conditions, not wrist-worn PPG sensors in general. A fair test of the placement hypothesis would require either two devices of comparable age and generation, or the same sensor model worn at both body sites.



(a) *Route comparison, Trial 3 (mean offset 8.26 m, max. 22.9 m for this run)*



(b) *Post-hoc power analysis ($\Delta = 5$ bpm, $\sigma_d = 6$ bpm, $\alpha = 0.05$)*

Figure 5: *Geospatial tracking accuracy for Trial 3 (a) and statistical power validation (b). $N = 6$ achieves 53.2% power; $n \approx 13$ trials would be required for the conventional 80% target.*

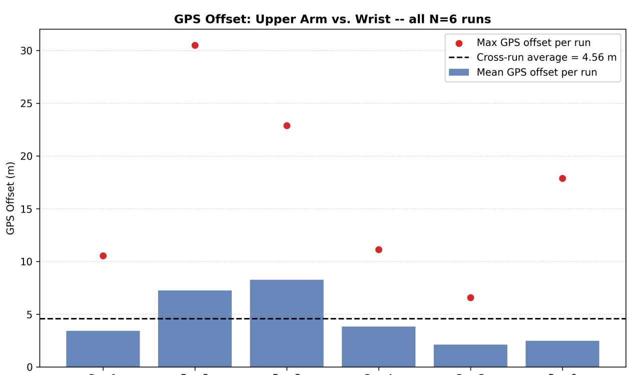


Figure 6: *GPS offset between upper-arm and wrist tracks across all $N = 6$ runs. The cross-run mean (4.56 m, dashed line) is the representative figure reported in the Abstract; per-run means range from 2.1 to 8.3 m, with individual maxima up to 30.5 m (Run 2).*

Agreement, not accuracy. This study measures agreement between two consumer-grade optical sensors, neither of which is a clinically validated reference. We use the term *accuracy* cautiously throughout; strictly speaking, our data support claims about *cross-device agreement and relative data density*, not accuracy against ground truth. A stronger design would add a third, independently validated reference device (e.g., a chest-worn ECG strap) to establish true accuracy for both optical sensors.

External validity. All $N = 6$ runs were performed by a single subject (intra-subject design). While this controls for inter-individual physiological variance, it means the reported statistics cannot be generalized to other runners, body compositions, or skin tones — factors known to affect PPG signal quality. Future work should extend this protocol across multiple participants and, ideally,

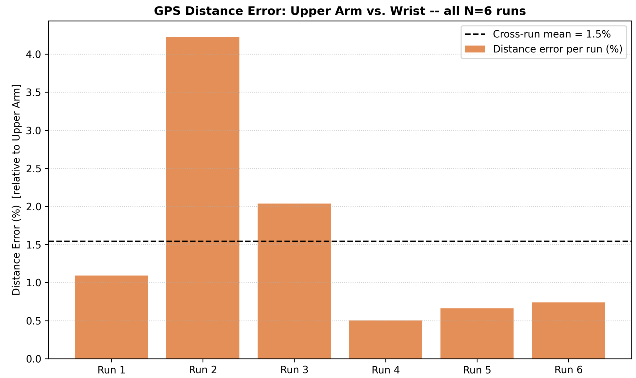
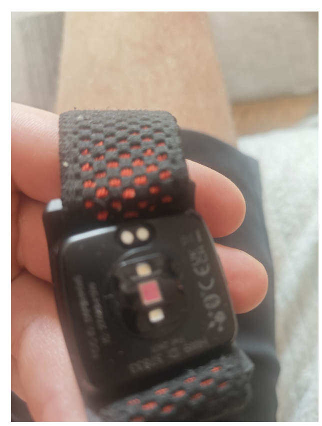


Figure 7: *Relative GPS track-length error (upper arm as reference) across all $N = 6$ runs. The cross-run mean (1.5%, dashed line) is again driven up by Run 2 (4.2%), consistent with the GPS-offset and HR-agreement outliers observed for the same run (see Section 5).*

Table 2: *Cross-run correlations ($N = 6$ runs treated as observations) between degradation metrics. Distance error and HR lag both correlate significantly with weaker HR agreement, supporting a shared underlying cause; speed agreement correlates with none of them, suggesting it fails independently.*

Metric pair	Pearson r	p
Distance error vs. HR r	-0.89	0.018
Distance error vs. HR MAE	$+0.84$	0.035
HR lag vs. HR r	-0.84	0.038
GPS mean offset vs. distance error	$+0.77$	0.075
GPS max offset vs. HR r	-0.73	0.100
GPS mean offset vs. HR r	-0.59	0.215
Speed r vs. HR r	-0.18	0.727
Speed r vs. distance error	-0.11	0.840
Speed r vs. GPS max offset	-0.05	0.922

(a) *Amazfit GTR 3 Pro (wrist sensor)*



(b) *Coospo HW9 (upper-arm sensor)*

Figure 8: *Visual comparison of the optical sensor arrays. (a) The over-4-year-old smartwatch shows a distinct round, milky-white discoloration and scratching directly over the PPG window. (b) The newer upper-arm array shows an intact, clear optical barrier.*

multiple device units per condition to separate placement effects from unit-specific wear. Geospatial tracking performed well overall in our data, with a study-wide mean spatial offset of ≈4.56 m across all six runs (range 2.1–8.3 m per-run mean, max. 30.5 m; Figure 6), though this combined average should not obscure the per-run variability visible in Table 1.

Outlook: a derived metric for practitioners. Independently of the device-comparison question above, amateur runners can make practical use of the acquired data via the Heart Rate Efficiency (HRE) metric, defined as the quotient of velocity (v) and heart rate (HR):

$$\mathrm{HRE} = \frac{v}{\mathrm{HR}} \tag{2}$$

This metric has been proposed to scale with aerobic adaptation, offering a potentially more stable feedback signal than raw heart rate tracking alone [Votyakov et al. 2025]. We flag it here only as a practical outlook rather than a validated contribution of this study: the formulation is currently supported by a single non-peer-reviewed preprint, and applying it would inherit all of the cross-device agreement limitations discussed above; we therefore recommend peer-reviewed replication before wider adoption.

6 Conclusion

Within this single-subject, six-trial pilot, mean heart rate and geospatial tracking agreed closely between the wrist-worn Amazfit GTR 3 Pro and the upper-arm Coospo HW9, but the aged wrist device showed substantial cardiac data loss (up to 74.6% in the worst runs) and missed several transient load peaks that the upper-arm device captured. Because the two devices differ simultaneously in placement, hardware generation, and physical condition, we cannot attribute this gap to sensor placement alone; a confirmatory follow-up should use generation-matched devices (or the same model at both sites) together with an independent reference such as a chest-worn ECG strap, and should recruit enough trials or participants to reach adequate statistical power ($n \approx 13$ trials indicated here). With those caveats, our data are consistent with upper-arm placement being a promising direction for high-density cardiac monitoring during intensive training, while wrist-worn tracking may remain adequate for steadier-state activity.

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References

CHOW, H.-W., AND YANG, C.-C. 2020. Accuracy of Optical Heart Rate Sensing Technology in Wearable Fitness Trackers for Young and Older Adults: Validation and Comparison Study. *JMIR mHealth and uHealth 8*, 4 (Apr.), e14707.

FULLER, D., COLWELL, E., LOW, J., ORYCHOCK, K., TOBIN, M. A., SIMANGO, B., BUOTE, R., VAN HEERDEN, D., LUAN, H., CULLEN, K., SLADE, L., AND TAYLOR, N. G. A. 2020. Reliability and Validity of Commercially Available Wearable Devices for Measuring Steps, Energy Expenditure, and Heart Rate: Systematic Review. *JMIR mHealth and uHealth 8*, 9 (Sept.), e18694.

GERMINI, F., NORONHA, N., BORG DEBONO, V., ABRAHAM PHILIP, B., PETE, D., NAVARRO, T., KEEPANASSERIL, A., PARPIA, S., DE WIT, K., AND IORIO, A. 2022. Accuracy and Acceptability of Wrist-Wearable Activity-Tracking Devices: Systematic Review of the Literature. *Journal of Medical Internet Research 24*, 1 (Jan.), e30791.

JACKO, T., BARTSCH, J., VON DIECKEN, C., AND UEBERSCHÄR, O. 2024. Validity of Current Smartwatches for Triathlon Training: How Accurate Are Heart Rate, Distance, and Swimming Readings? *Sensors 24*, 14 (July), 4675.

KIM, K. B., AND BAEK, H. J. 2023. Photoplethysmography in Wearable Devices: A Comprehensive Review of Technological Advances, Current Challenges, and Future Directions. *Electronics 12*, 13 (July), 2923.

POBIRUCHIN, M., SULEDER, J., ZOWALLA, R., AND WIESNER, M. 2017. Accuracy and adoption of wearable technology used by active citizens: A marathon event field study. *JMIR mHealth and uHealth 5*, 2, e24.

SCHWEIZER, T., AND GILGEN-AMMANN, R. 2024. Heart rate measurement using optical technology on the arm and wrist during different activities at different intensities: An analytical validation study. *JMIR Preprints*.

STAHL, S. E., AN, H.-S., DINKEL, D. M., NOBLE, J. M., AND LEE, J.-M. 2016. How accurate are the wrist-based heart rate monitors during walking and running activities? Are they accurate enough? *BMJ Open Sport & Exercise Medicine 2*, 1 (Apr.), e000106.

VOTYAKOV, A., ET AL., 2025. A longitudinal evaluation of heart rate efficiency for amateur runners. arXiv:2509.05961v2.